Quan Nguyen

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University of Florida

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Research Interests

AI Security, Distributed Machine Learning, Multi-Agent AI, Quantum Computing.

Federated Learning, Differential Privacy, Multi-Agent System Security, Adversarial Machine Learning, Foundation Models, Quantum Machine Learning.

Education

2025 - now	**Ph.D.**	University of Florida, PhD Student, Computer Science.
2024	**M.Sc.**	Technical University of Munich, M.Sc. Biomedical Computing.
2022	**B.Sc.**	Hanoi University of Science and Technology, B.Sc. Information Technology
2021	**B.Eng.**	Hanoi University of Science and Technology, B.Eng. Biomedical Engineering

Experience

May - Aug 2026 **Ph.D. Research Intern** - Data Trust Team **Seagate Technology, U.S**.

- **Neuron Alignment for Federated Model Merging:** Developed a Fisher-weighted optimal-transport method that aligns neurons across client models and merges them according to each parameter's importance, for one-shot and few-round privacy-preserving federated learning under strong data heterogeneity.

2025 - now **Research Assistant** - Adaptive Learning and Optimization Lab **University of Florida, U.S**.

- **AI Security:** Develop novel privacy attacks and differentially private mechanisms to mitigate adversarial privacy risks in Classical Deep Learning and Quantum Machine Learning.
  [NeurIPS '26, AISTATS '26, ICML '25]

- **Decentralized Foundation Models:** Work on decentralized training and parameter-efficient fine-tuning of Foundation Models (Large Language Models, Vision Transformers) in Heterogeneous Environments.

- **Explainable Federated Learning:** Develop neuron provenance methods that trace neurons in the aggregated global model back to the clients that contributed them, explaining how each client shapes the merged model. *Work in collaboration with the Data Trust Team, Seagate Technology.*

- **Multi-Agent AI Security:** Investigate cross-layer security risks across multiple channels (e.g., communication, memory) and cascading failures in Multi-Agent Systems.

2024 **Master's Thesis Student** - AI in Medicine Lab **TU Munich/ Siemens Healthineers, Germany**

- **Differentially Private Deep Learning:** Investigated the disparate impact of differential privacy on AI models. Identified algorithmic encoding of protected patient characteristics through shortcut learning. Developed novel clipping and noise-addition mechanisms for DP-SGD.

2023 - 2024 **Internship** - Computer Vision and Algorithmics Team **LUMA Vision GmbH, Germany**

- **Deep Learning for Ultrasound Imaging:** Built an end-to-end deep learning framework for 4D intracardiac ultrasound image segmentation. Implemented a deep learning-based speckle filtering model and SVD clutter filtering to enhance image quality. Deployed on a real-world product. **[IEEE IUS '24]**

2022 **Research Assistant** - Security and Artificial Intelligence Lab **VinUniversity, Vietnam**

- **Federated Learning on Edge Devices:** Developed a federated learning algorithm that decomposes a large model into an ensemble of lightweight sub-models, enabling clients to train them in parallel across multiple devices without sharing data via split learning. This approach achieved 4-8× reduction in client memory usage while preserving privacy and incurring no additional server overhead. **[IEEE TNSM '23]**

2021 **Computer Vision Engineer** - AI Camera Team **Viettel High Tech, Vietnam**

- **AI on Edge Devices:** Developed face recognition algorithms on edge devices for the **NIST Face Recognition Technology Evaluation** (#1 Vietnamese vendor in BORDER category at submission). Developed real-time AI detection for fisheye and traffic cameras. [**IEEE AVSS '21 & ICISN '21**]

Selected Publications

1. Conference **[NeurIPS '26] Quan Minh Nguyen**, Min-Seon K., Hoang M.N, Trong N.H, Hyuk-Yoon K., My T. Thai. "Leveraging Soft Prompts for Privacy Attacks in Federated Prompt Tuning," Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS). A* Conference.

2. Conference **[AISTATS '26]** Hoang M.N, Nhat H.X, **Quan Minh Nguyen**, Nguyen H. K. D., Incheol Shin, My T. Thai, "Q-ShiftDP: A Differentially Private Parameter-Shift Rule for Quantum Machine Learning," Twenty-Ninth Annual Conference on Artificial Intelligence and Statistics (AISTATS). A* Conference.

3. Conference **[ICML '25] Quan Minh Nguyen***, Minh N. Vu*, Truc Nguyen, My T. Thai. Theoretically Unmasking Inference Attacks Against LDP-Protected Clients in Federated Vision Models. Forty-Second International Conference on Machine Learning (ICML). A* Conference.

4. Conference **[IEEE IUS '24]** Martina Casagrande, **Quan Nguyen**, Ivan Dudurych, Christoph Hennersperger, Stefan Wörz. Advancing 3D ICE: Challenges and Deep Learning Strategies in Atrium Segmentation. IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium. Flagship conference of IEEE UFFC Society.

5. Journal **[IEEE TNSM '23] Nguyen, Q**., Pham, H. H., Wong, K. S., Le Nguyen, P., Nguyen, T. T., & Do, M. N. (2023). FedDCT: Federated Learning of Large Convolutional Neural Networks on Resource Constrained Devices Using Divide and Collaborative Training. IEEE Transactions on Network and Service Management. Q1 Journal.

6. Conference [**IEEE AVSS '21**] **Minh, Q. N**., Le Van, B., Nguyen, C., Le, A., & Nguyen, V. D. (2021, November). ARPD: Anchor-free Rotation-aware People Detection using Topview Fisheye Camera. In 2021 17th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS).

7. Conference **[ICISN '21]** Nguyen, V. D., Pham, P. N., Nguyen, X. B., Tran, T. M., & **Nguyen, M. Q.** (2021, March). Incorporation of panoramic view in fall detection using omnidirectional camera. In The International Conference on Intelligent Systems & Networks.

Technical Skills

- **Programming Languages:** C, C++, Python, Java, Gleam.
- **Libraries:** PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, SciPy, Opacus, JAX Privacy, OpenCV, Wandb, Flower, CuPy, Hugging Face, Matplotlib, Seaborn, ONNX, TensorRT.
- **Systems/ Tools:** Git, LaTeX, HPC, Linux, Docker, Slurm, Google Cloud.

Activities

Awards
CISE Graduate Scholarship at University of Florida (2026)
Scholarship - The 9th Vietnam Summer School of Science (2023)
Scholarship - "Modern machine learning: Foundations and applications" Program. (2023)
First prize in Vietnam AI Day 2022 - with solution: "VAIPE: Protective healthcare monitoring and supporting system for Vietnamese" (2022)
Third prize in Hanoi University of Science & Tech **Academic Research Contest** (2021)
Scholarship - Hanoi University of Science & Tech students with **top academic results** (2020)
Scholarship - FPT Center for Young Talents (25 awarded nationwide per year) (2019)
International Exchange Scholarship at Temasek Polytechnic, Singapore (2018)

Teaching COMP 1020 - OOP & Data Structures - VinUni (S'22), CDA 3101 - Comp. Org. - UF (S'26).

Service **Organizer & Webmaster**: CSoNet 2025–26; **Workshop Organizer:** VinUni-Illinois Smart Health 2022; **PC Member:** SOICT 2023; **Reviewer:** IEEE AVSS, AISTATS, IEEE Access, IEEE Trans. Dependable & Secure Computing, ACM Computing Surveys, ACM HEALTH, TMLR.

Languages **English (IELTS 8.5, C2 Level)**, German (A2), Vietnamese (Native).

References

1. **Prof. My T. Thai**, UF Research Foundation Professor of Computer & Information Sciences & Engineering and Associate Director of Warren B. Nelms Institute, IEEE Fellow, University of Florida. mythai@cise.ufl.edu
2. Dr. Stefan Wörz, Head of Computer Vision and Algorithmics, LUMA Vision. stefan.woerz@lumavision.com
3. Dr. Yasaman Keshtkarjahromi, Staff Engineer, Seagate Technology. yasaman.keshtkarjahromi@seagate.com